Section 3.1 - Limits

Definitions:

Left-Hand Limit: We write $\lim_{x \to a^-} f(x) = K$ if as x approaches a from the left, $f(x)$ approaches K.

"the limit of f(x) as x approaches a from the left is K"

*we only care about what is going on near a and not at a.

Right-Hand Limit: We write $\lim_{x \to a^+} f(x) = L$ if as x approaches a from the right, $f(x)$ approaches L.

"the limit of f(x) as x approaches a from the right is L"

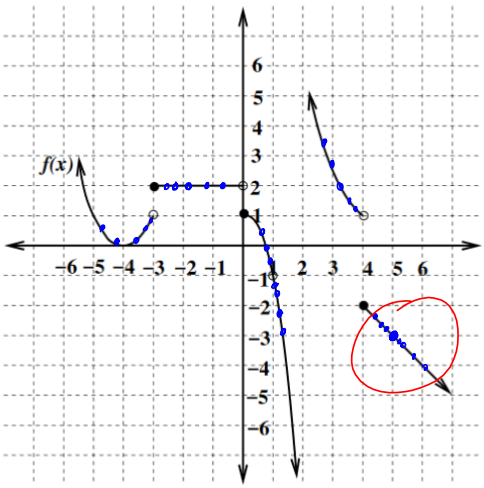
(Two-Sided) Limit: We write $\lim_{x \to a} f(x) = M$ if the functional value $f(x)$ approaches M as x approaches a (from either side of a).

"the limit of f(x) as x approaches a is M"

Note: For a (two-sided) limit to exist, the limit from the left and the limit from the right must both exist and be equal. That is

$\lim_{x \to a} f(x) = M$ if and only if $\lim_{x \to a^-} f(x) = M$ and $\lim_{x \to a^+} f(x) = M$

Example 1: Use the graph below to find the following limits:



$\lim_{x \to 5} f(x) = -3 = f(5)$

a) $\lim_{x \to -3^-} f(x) = \boxed{1}$

b) $\lim_{x \to -3^+} f(x) = \boxed{2}$

c) $\lim_{x \to -3} f(x)$ $\boxed{\text{Does Not Exist}}$ $\boxed{\text{DNE}}$

d) $\lim_{x \to 1^-} f(x) = \boxed{-1}$

e) $\lim_{x \to 1^+} f(x) = \boxed{-1}$

f) $\lim_{x \to 1} f(x) = \boxed{-1}$

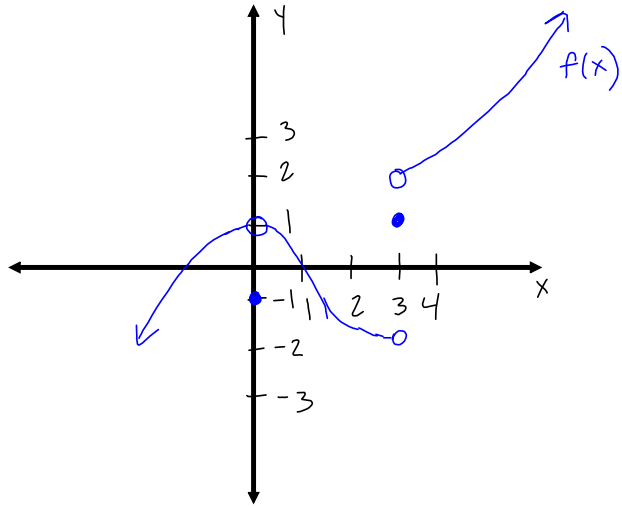
g) $\lim_{x \to 4^-} f(x) = \boxed{1}$

h) $\lim_{x \to 4^+} f(x) = \boxed{-2}$

i) $\lim_{x \to 4} f(x)$ $\boxed{\text{DNE}}$

Example 2: Sketch the graph of an example of a function f that satisfies all of the given conditions.

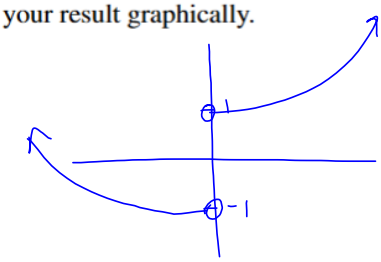
- $\lim_{x\to 0} f(x) = 1$
- $\lim_{x\to 3^-} f(x) = -2$
- $\lim_{x\to 3^+} f(x) = 2$
- $f(0) = -1$, $f(3) = 1$



Example 3: Use a table of values to estimate $\lim_{x\to 0} \frac{\overbrace{x^2+x}^{f(x)}}{\sqrt{x^3+x^2}}$. Confirm your result graphically.

x	$f(x)$
$-.1$	$-.948683$
$-.01$	$-.994987$
$-.001$	$-.999500$

x	$f(x)$
$.1$	1.048809
$.01$	1.004988
$.001$	1.000500



$\lim_{x\to 0^-} f(x) = -1$ $\lim_{x\to 0^+} f(x) = 1$

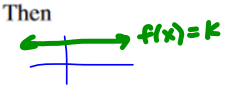
Thus, $\boxed{\lim_{x\to 0} f(x) \text{ does not exist.}}$

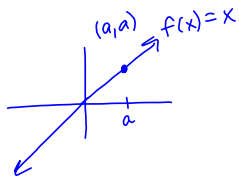
Limits Laws

Let f and g be two functions, and assume that

$$\lim_{x\to a} f(x) = L \qquad \lim_{x\to a} g(x) = M$$

where L and M are real numbers (both limits exist). Then





1. $\lim_{x\to a} k = k$ for any constant k
2. $\lim_{x\to a} x = a$
3. $\lim_{x\to a}[f(x)+g(x)] = \lim_{x\to a} f(x) + \lim_{x\to a} g(x) = L+M$
4. $\lim_{x\to a}[f(x)-g(x)] = \lim_{x\to a} f(x) - \lim_{x\to a} g(x) = L-M$
5. $\lim_{x\to a} kf(x) = k\lim_{x\to a} f(x) = kL$ for any constant k
6. $\lim_{x\to a}[f(x)\cdot g(x)] = [\lim_{x\to a} f(x)][\lim_{x\to a} g(x)] = LM$
7. $\lim_{x\to a} \frac{f(x)}{g(x)} = \frac{\lim_{x\to a} f(x)}{\lim_{x\to a} g(x)} = \frac{L}{M}$ if $M \neq 0$
8. $\lim_{x\to a}[f(x)]^n = \left[\lim_{x\to a} f(x)\right]^n$ where n is a postive integer
9. $\lim_{x\to a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x\to a} f(x)} = \sqrt[n]{L}$, $L > 0$ for n even

Example 4: Determine the following limits:

a) $\lim_{x\to 2} 10 = \boxed{10}$

b) $\lim_{x\to 3} x^2+3x-2 = \lim_{x\to 3} x^2 + \lim_{x\to 3} 3x - \lim_{x\to 3} 2 = \left(\lim_{x\to 3} x\right)^2 + 3\lim_{x\to 3} x - 2$

$= (3)^2 + 3(3) - 2 = \boxed{16}$

c) $\lim_{x\to 3} \frac{3x+2}{x-4} = \frac{\lim_{x\to 3}(3x+2)}{\lim_{x\to 3}(x-4)} = \frac{3\cdot\lim_{x\to 3} x + \lim_{x\to 3} 2}{\lim_{x\to 3} x - \lim_{x\to 3} 4} = \frac{3(3)+2}{3-4} = \frac{11}{-1} = \boxed{-11}$

Limits of Polynomial and Rational Functions

1. $\lim_{x\to a} f(x) = f(a)$, f any polynomial function
2. $\lim_{x\to a} r(x) = r(a)$, r any rational function with nonzero denominator at $x = a$

Example 5: Find each limit.

a) $\lim_{x\to 3} x^3 - 4x^2 + 2 = 3^3 - 4(3)^2 + 2 = \boxed{-7}$

b) $\lim_{x\to 2} \frac{x^2-1}{x+2} = \frac{2^2-1}{2+2} = \boxed{\frac{3}{4}}$

c) If $f(x) = \begin{cases} x^2+3 & \text{if } x \le 3 \\ x-4 & \text{if } x > 3 \end{cases}$ find

i) $\lim_{x\to 0} f(x) = \lim_{x\to 0} (x^2+3) = 0^2+3 = \boxed{3}$

(ii) $\lim_{x\to 3^-} f(x) = \lim_{x\to 3^-} (x^2+3) = 3^2+3 = \boxed{12}$

(iii) $\lim_{x\to 3^+} f(x) = \lim_{x\to 3^+} (x-4) = 3-4 = \boxed{-1}$

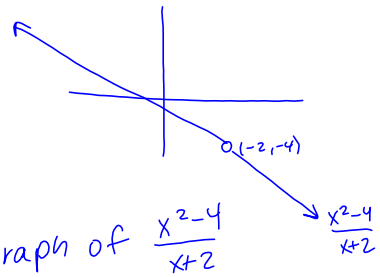
(iv) $\lim_{x\to 3} f(x)$ $\boxed{\text{Does Not Exist}}$

d) $\lim_{x\to -2} \frac{x^2-4}{x+2} = \frac{(-2)^2-4}{-2+2} = \frac{0}{0}$ ← known as indeterminate form

$= \lim_{x\to -2} \frac{\cancel{(x+2)}(x-2)}{\cancel{(x+2)}}$

$= \lim_{x\to -2} (x-2)$

$= -2-2 = \boxed{-4}$



the graph of $\frac{x^2-4}{x+2}$ and $x-2$ are exactly the same for all values of x except $x=-2$.

How to find limits algebraically $\left(\lim_{x\to a} f(x)\right)$

Try plugging a into the function:

① If you get a number that's your answer (unless you are dealing with a piecewise function)

② If you get $\frac{0}{0}$ (indeterminate form), algebraically manipulate (usually factor), cancel, & plug in a again.

③ If you get $\frac{\text{nonzero \#}}{0}$ then the limit does not exist.

Example 6: Evaluate each limit.

a) $\lim_{x\to 2} \frac{2x^2-x-6}{x-2} = \frac{2(2)^2-2-6}{2-2} = \frac{0}{0}$

$= \lim_{x\to 2} \frac{(2x+3)\cancel{(x-2)}}{\cancel{(x-2)}}$

$= \lim_{x\to 2} (2x+3) = 2(2)+3 = \boxed{7}$

b) $\lim_{x\to 5} \frac{|x-5|}{x-5} = \frac{|5-5|}{5-5} = \frac{0}{0}$

Recall, $|x-5| = \begin{cases} -(x-5), & x<5 \\ x-5, & x\geq 5 \end{cases}$

$\frac{|x-5|}{x-5} = \begin{cases} \frac{-\cancel{(x-5)}}{\cancel{x-5}} = -1, & x<5 \\ \frac{x-5}{x-5} = 1, & x\geq 5 \end{cases}$

$\lim_{x\to 5^-} \frac{|x-5|}{x-5} = \lim_{x\to 5^-} -1 = -1$

$\lim_{x\to 5^+} \frac{|x-5|}{x-5} = \lim_{x\to 5^+} 1 = 1$

$\lim_{x\to 5} \frac{|x-5|}{x-5}$ $\boxed{\text{DNE}}$

c) $\lim_{x\to -4} \frac{\sqrt{x^2+9}-5}{x+4} = \frac{\sqrt{(-4)^2+9}-5}{-4+4} = \frac{0}{0}$

$= \lim_{x\to -4} \frac{(\sqrt{x^2+9}-5)}{x+4} \frac{(\sqrt{x^2+9}+5)}{(\sqrt{x^2+9}+5)}$

$(x^2+9)^{1/2}(x^2+9)^{1/2} = (x^2+9)^{1/2+1/2} = x^2+9$

$(\sqrt{x^2+9}-5)(\sqrt{x^2+9}+5)$

$= \sqrt{x^2+9}\sqrt{x^2+9} + \cancel{5\sqrt{x^2+9}} - \cancel{5\sqrt{x^2+9}} - 25$

$= x^2+9-25$

$= x^2-16$

$= \lim_{x\to -4} \frac{x^2-16}{(x+4)(\sqrt{x^2+9}+5)} = \lim_{x\to -4} \frac{\cancel{(x+4)}(x-4)}{\cancel{(x+4)}(\sqrt{x^2+9}+5)}$

$= \lim_{x\to -4} \frac{x-4}{\sqrt{x^2+9}+5} = \frac{-4-4}{\sqrt{(-4)^2+9}+5} = \frac{-8}{5+5} = \frac{-8}{10} = \boxed{\frac{-4}{5}}$

d) If $f(x) = 3x^2+4x-7$ find $\lim_{h\to 0} \frac{f(x+h)-f(x)}{h} = \frac{f(x)-f(x)}{0} = \frac{0}{0}$

$\lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$

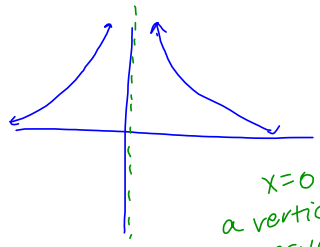
$f(x+h) = 3(x+h)^2+4(x+h)-7$
$= 3(x+h)(x+h)+4x+4h-7$
$= 3(x^2+2xh+h^2)+4x+4h-7$
$= 3x^2+6xh+3h^2+4x+4h-7$

$= \lim_{h\to 0} \frac{\cancel{3x^2}+6xh+3h^2+\cancel{4x}+4h\cancel{-7}-\cancel{3x^2}-\cancel{4x}\cancel{+7}}{h}$

$= \lim_{h\to 0} \frac{6xh+3h^2+4h}{h}$

$= \lim_{h\to 0} \frac{\cancel{h}(6x+3h+4)}{\cancel{h}} = \lim_{h\to 0} (6x+3h+4) = 6x+3(0)+4 = \boxed{6x+4}$

e) $\lim_{x\to 0} \frac{7-x}{x^2}$ $= \frac{7-0}{0^2} = \frac{7}{0}$ $\boxed{\text{Does Not Exist}}$



x=0 is a vertical asymptote

Example 7: Let's examine the behavior of $f(x) = \frac{7-x}{x^2}$ at $x = 0$.

x	f(x)
-.01	70100
-.001	7001000
-.0001	700010000

$\lim_{x\to 0^-} f(x) = \infty$

x	f(x)
.01	69900
.001	6999000
.0001	699990000

$\lim_{x\to 0^+} f(x) = \infty$

Thus, $\boxed{\lim_{x\to 0} f(x) = \infty}$

this describes the way in which it does not exist.

better answer than just Does Not Exist.

Definition: The vertical line $x = a$ is a **vertical asymptote** for the graph of $y = f(x)$ if

$\lim_{x\to a^-} f(x) = \pm\infty$ or $\lim_{x\to a^+} f(x) = \pm\infty$

Locating Vertical Asymptotes of Rational Functions

If $f(x) = \frac{n(x)}{d(x)}$ is a rational function, $d(c) = 0$ and $n(c) \neq 0$, then the line $x = c$ is a vertical asymptote of the graph of f.

Example 8: Describe the behavior of f at each value of x that is not in its domain.

$f(x) = \frac{x-3}{x^2-4x+3}$

$x^2-4x+3 \neq 0$

$(x-1)(x-3) \neq 0$

$x \neq 1,\ x \neq 3$

x=1

$\lim_{x\to 1} \frac{x-3}{x^2-4x+3} = \frac{1-3}{1^2-4(1)+3} = \frac{-2}{0}$

$\boxed{\text{x=1 is a vertical asymptote}}$

x	f(x)
.9	-10
.99	-100
.999	-1000

$\lim_{x\to 1^-} f(x) = -\infty$

x	f(x)
1.1	10
1.01	100
1.001	1000

$\lim_{x\to 1^+} f(x) = \infty$

$\lim_{x\to 1} f(x)$ Does Not Exist

x=3

$\lim_{x\to 3} \frac{x-3}{x^2-4x+3} = \frac{3-3}{3^2-4(3)+3} = \frac{0}{0}$

$= \lim_{x\to 3} \frac{\cancel{(x-3)}}{\cancel{(x-3)}(x-1)} = \lim_{x\to 3} \frac{1}{x-1} = \frac{1}{3-1} = \frac{1}{2}$

we have a hole in the graph at (3, 1/2)

Example 9: Find the limit. Use $-\infty$ and ∞ when appropriate. (You try)

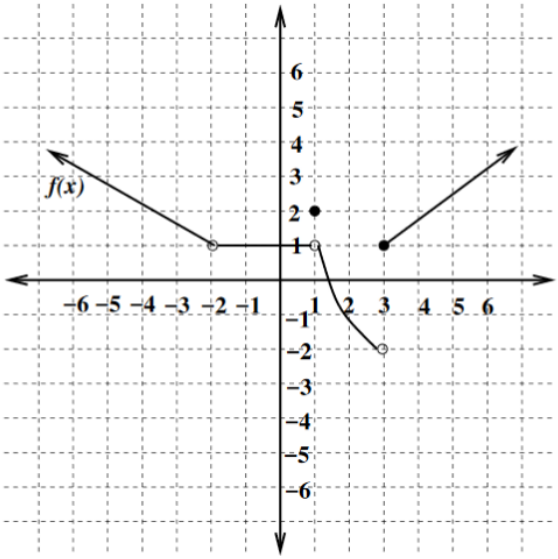
$\lim_{x\to -3} \frac{x^2}{x+3}$

Definition: A function f is **continuous** at the point $x = a$ if all of the following are true:

1. $f(a)$ exists
2. $\lim_{x \to a} f(x)$ exist.
3. $\lim_{x \to a} f(x) = f(a)$

Note: If one or more of the conditions are not met, we say f is discontinuous at $x = a$.

Example 10: For what values of x is the function discontinuous? Explain.



$x = -2$

$f(-2)$ does not exist

$x = 1$

$\lim_{x \to 1} f(x) \neq f(1)$

$x = 3$

$\lim_{x \to 3} f(x)$ does not exist

Definition: A function is **continuous on an interval** if it is continuous at each point on the interval.

The following types of functions are continuous at every number in their domains:

polynomials, exponentials, logarithmic, rationals, & root functions

polynomials: $\mathbb{R}$

exponentials: $\mathbb{R}$

logarithmic: inside > 0

rationals: denom $\neq 0$

root functions: odd, $\mathbb{R}$; even, inside ≥ 0

Example 11: Find the intervals on which the following functions are continuous.

a) $f(x) = 3x^9 + 4x^5 - 7$ poly $\Rightarrow$ $\boxed{(-\infty, \infty)}$

b) $g(x) = 2^{3x} + 4$ exp $\Rightarrow$ $\boxed{(-\infty, \infty)}$

c) $h(x) = \log_8(x - 3) - 2$ log $\Rightarrow$ inside > 0

$x - 3 > 0$

$x > 3$

$\boxed{(3, \infty)}$

d) $k(x) = \dfrac{x+2}{x^2-3x-10}$

rational $\Rightarrow$ $x^2-3x-10 \neq 0$

$(x+2)(x-5) \neq 0$

$x+2 \neq 0$ $\quad$ $x-5 \neq 0$

$x \neq -2$ $\quad$ $x \neq 5$

$\boxed{(-\infty,-2)\cup(-2,5)\cup(5,\infty)}$

e) $m(x) = \begin{cases} \dfrac{x-5}{x+2} & \text{if } x \leq -3 \\ \dfrac{x}{x-4} & \text{if } x > -3 \end{cases}$

Note: we have a formula for all values of x

Let's look at where we are switching from branch to branch: $x=-3$

1st branch: $\dfrac{x-5}{x+2}$

~~discont. at $x=-2$~~

not on the interval $x \leq -3$ so not a problem for $m(x)$

2nd branch: $\dfrac{x}{x-4}$

discont. at $x=4$

$m(x)$ is discont. at $x=4$ since its on the interval $x>-3$

$m(-3) = \dfrac{-3-5}{-3+2} = \dfrac{-8}{-1} = 8$ ✓

$\lim\limits_{x\to -3^-} m(x) = \lim\limits_{x\to -3^-} \dfrac{x-5}{x+2} = \dfrac{-3-5}{-3+2} = \dfrac{-8}{-1} = 8$

$\lim\limits_{x\to -3^+} m(x) = \lim\limits_{x\to -3^+} \dfrac{x}{x-4} = \dfrac{-3}{-3-4} = \dfrac{-3}{-7} = \dfrac{3}{7}$

Thus, $\lim\limits_{x\to -3} m(x)$ Does Not Exist

$\Rightarrow$ $m(x)$ is discont. at $x=-3$

$\boxed{(-\infty,-3)\cup(-3,4)\cup(4,\infty)}$

Example 12: Find the value(s) of k that make $f(x)$ continuous everywhere.

$f(x) = \begin{cases} 2x-5 & \text{if } x \leq 1 \\ x^2+k & \text{if } x > 1 \end{cases}$

- first notice its defined for all values of x ✓
- there are no issues on each branch since they are both poly ✓

$f(1) = 2(1)-5 = -3$ ✓

$\lim\limits_{x\to 1^-} f(x) = \lim\limits_{x\to 1^-} 2x-5 = 2(1)-5 = -3$

$\lim\limits_{x\to 1^+} f(x) = \lim\limits_{x\to 1^+} x^2+k = (1)^2+k = 1+k$

$\lim\limits_{x\to 1^-} f(x) = \lim\limits_{x\to 1^+} f(x)$

$-3 = 1+k$

$-1 \quad -1$

$\boxed{-4 = k}$

Section 3.1 Highly Suggested Homework Problems: 1-45(odd), 49, 51, 61-81(odd)